

From Data to Dialogue – AI Insights for Bank Employees

How can 10,000 customer records with transactions be transformed into a machine-learning pipeline that delivers relevant, understandable insights to bank employees?

YOUR CHALLENGE

Using 10,000 synthetic customer records incl. transactions, build an ML pipeline and frontend that delivers AI insights to bank employees – focusing on usability and value for both bank and customer.

CONTEXT

Banks hold valuable data: customer profiles, account movements and transaction patterns. But raw data alone helps no one – insights create value: Which customer may soon need a conversation? Where are untapped opportunities? Which anomalies deserve attention?

The challenge is not only the model, but the entire chain: from data preparation and the ML model to a clear presentation in day-to-day work. An insight that nobody understands or trusts is worthless.

You will receive 10,000 synthetic customer records including transactions – no real customer data, but realistic patterns.

USER QUESTIONS

- Which of my customers should I contact this week – and why?
- Which patterns (saving behaviour, spending, life events) are relevant?
- How can I identify opportunities and risks early?
- Why is the system suggesting this specific insight (explainability)?
- How does the insight feed into my customer conversation?
- What does the customer gain from it?

DESIGN PRINCIPLE

Not: “A dashboard full of charts.”

Instead: “A few insights that are easy to understand and lead to action.”

THE PIPELINE

Ingest Load and understand data

Prepare Build features from transactions and profiles

Model Segmentation, prediction, anomalies, next best action

Explain Explain insights and make them understandable

Serve Surface insights in the employee workflow

Feedback Employees rate insights, model learns

POSSIBLE MVP COMPONENTS

- Working ML pipeline (even simplified) on the dataset
- Customer segmentation or behavioural clusters
- 2–3 concrete insight types (e.g. savings potential, churn signal, life event)
- Insight feed or customer card for the frontend
- Explainability: “Why am I seeing this?”
- Feedback loop (insight useful / not useful)

WHY HACK?

- You work with realistic data: 10,000 customers incl. transactions.
- You build end-to-end: from raw data to the model and frontend.
- You show how ML creates real business value – not just accuracy.
- Explainability and usability: crucial AI skills in banking.

GUIDING QUESTION

How can we enable a bank employee to understand the relevant insights for each customer in under one minute – and know what to do with them?

NORTH STAR

Insight to Action

WHAT WE WANT TO SEE AT THE END

Not a perfect model with maximum accuracy. Use a working pipeline and a simple frontend to show how raw data becomes understandable, actionable insights – with clear value for both bank and customer.

Do not generate as many insights as possible – create as much impact as possible in customer interactions.

Basis: 10,000 synthetic customer records incl. transactions